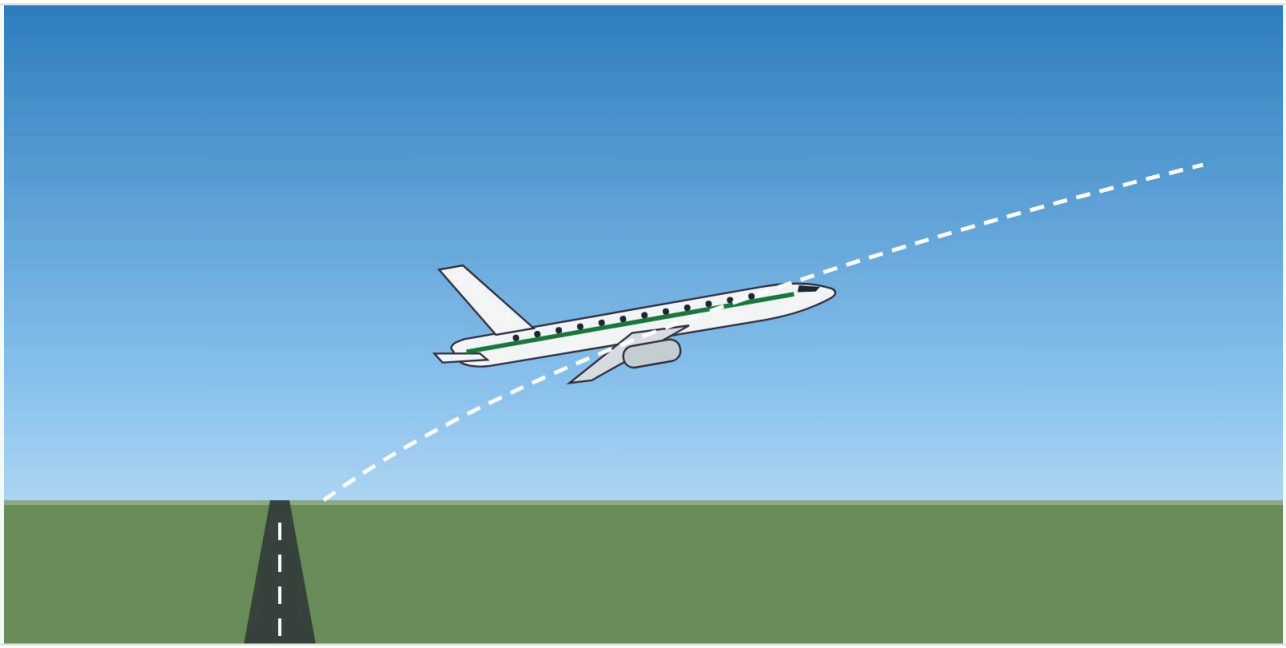
MODULE 2 · PBN OPERATIONS

# Lesson 03: RNAV Departures, Arrivals and En-route Operations

Lesson 3 of 5 · about 6 min

**LEARNING OBJECTIVES**

- Fly RNAV SIDs and STARs correctly.
- Apply flight technical error limits.
- Manage contingencies on RNAV routes.



Fly the SID with LNAV and respect the constraints.

## Before departure

- Check that the aircraft and crew are approved and the equipment required by the MEL is operational.
- Load and cross-check the SID: waypoints, fly-by and fly-over points, altitude and speed constraints.
- Brief the procedure and any changes that ATC may give (e.g. direct-to clearances).

## Flying the procedure

- Use the lateral navigation mode (LNAV) and the flight director or autopilot, as the SOP requires, to keep the **flight technical error** small — typically within half the navigation accuracy (for RNAV 1 / RNP 1: within 0.5 NM).
- Monitor the cross-track deviation and the navigation performance indications.
- Respect the published altitude and speed constraints — they are part of the procedure.

## Contingencies

- If the navigation performance cannot be met (alert, loss of GNSS, system failure), **inform ATC** — “unable RNAV due equipment” or the appropriate phrase — and request an alternative clearance.
- Follow the QRH / FCOM procedure for the navigation failure.
- Report the event.

**KEY TAKEAWAYS**

- Load and cross-check the SID or STAR, including constraints.
- Keep flight technical error within half the navigation accuracy.
- If performance is lost, tell ATC and request another clearance.

**KNOWLEDGE CHECK**

**On an RNAV 1 SID, the lateral flight technical error should normally be kept within:**

A. 0.1 NM

B. 0.5 NM

C. 2 NM

D. 5 NM

**Answer: B.** Typically within half the navigation accuracy: 0.5 NM for RNAV 1.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.